EXPERIMENT NO: 19

Experiment to understand Logistic Regression for a given dataset.

PROGRAM:

```
import numpy as np
import pandas as pd
df=pd.read_csv('Social_Network_Ads.csv')
print(df)
print(df.head())
features=df.iloc[:,[2,3]].values
label=df.iloc[:,4].values
print(features)
print(label)
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
for i in range(1,401):
  x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_state=42)
  model=LogisticRegression()
  model.fit(x_train,y_train)
  train_score=model.score(x_train,y_train)
  test_score=model.score(x_test,y_test)
  if test_score>train_score:
      print("Test {} Train {} Random State {}".format(test_score,train_score,i))
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_state=42)
finalModel=LogisticRegression()
finalModel.fit(x_train,y_train)
print(finalModel.score(x_train,y_train))
print(finalModel.score(x_test,y_test))
from sklearn.metrics import classification_report
print(classification_report(label,finalModel.predict(features)))
```

OUTPUT:

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array([[       19,  19000],
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[35,20000],[26,43000],[27,57000],[19,76000],[45,101000],[39,59000],[46,72000],[28,82000],[36,46000],[33,58000],[42,142000],[29,90000][29,92000],[25,88000],[40,99000],[38,79000],[33,48000],[28,102000],[44,94000],[33,99000],[28,86000],[33,75000],[31,57000],[43,5400],
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